Thadomal Shahani Engineering College, Bandra
Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha				Subject: ASD&D
TE - 1	Batch: T13	Sem: 5	Roll No: 54	Date:03/09/2026

Experiment 5

Aim: To automate application deployment using GitHub/Jenkins.

LO mapped: LO3

LO: Build and manage automated CI/CD pipelines for application development, testing, packaging and deployment.

Theory:

Automating Application Deployment using GitHub and Jenkins

In software development, writing code is only the first step. After writing code, developers need to test it and put it on a live server so people can use it. Doing this manually every time takes a lot of time and can cause human errors. To solve this problem, we use automation tools like Jenkins and GitHub.

Continuous Integration and Continuous Deployment (CI/CD) is a method to frequently deliver apps to customers by bringing automation into the stages of app development.

GitHub is a platform where developers store, share, and manage their code online. It acts as a central safe place for team members to keep all their project files.

Jenkins is a free automation server. Its main job is to watch the code stored in GitHub. Whenever a developer uploads new code to GitHub, Jenkins automatically downloads that new code, builds the project, tests it to make sure there are no errors, and then deploys it to the final server.

How GitHub and Jenkins work together:

- Code Commit: A developer writes new code on their computer and pushes it to a GitHub repository.
- SCM Integration: Jenkins is connected to this GitHub repository using Source Code Management (SCM). Jenkins can use polling (checking GitHub every few minutes) or webhooks (GitHub tells Jenkins immediately) to know when new code is added.
- Pipeline Execution: Once Jenkins sees the new code, it starts a Pipeline. A Pipeline is a set of instructions written in a special file called a Jenkinsfile. This file is stored directly inside the GitHub repository along with the code.
- Pipeline Stages: The Jenkinsfile has different stages like Build, Test, and Deploy. Jenkins runs these stages step by step in the correct order.
- Success or Failure: If all stages pass without problems, the application is successfully deployed. If any stage fails, Jenkins stops the whole process and alerts the developer to fix the error.

Benefits of using this automation:

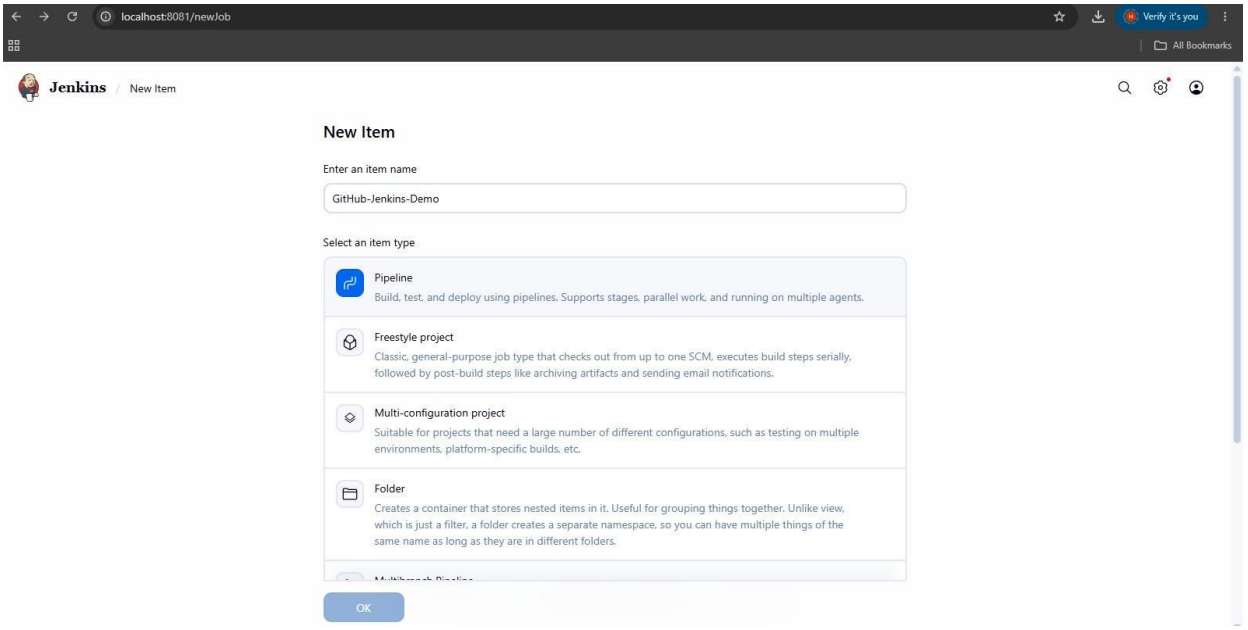
- Saves Time: Developers do not need to manually test and copy files to the server. The server

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does it for them.

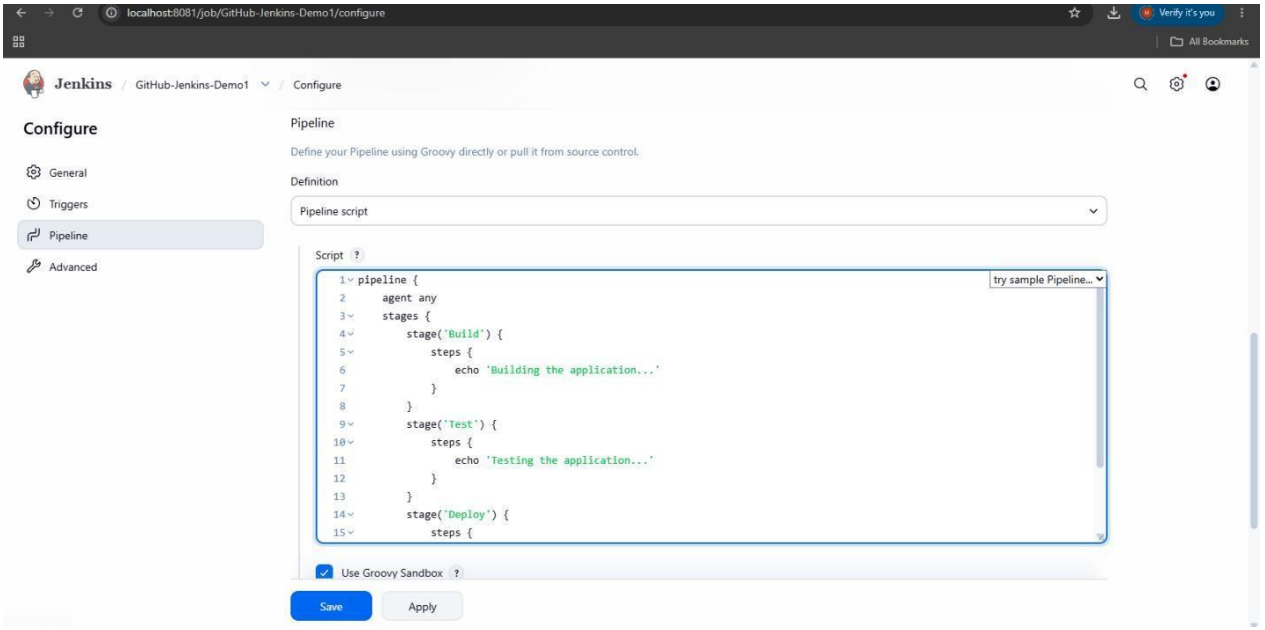
- Less Errors: Automated tests catch mistakes very early before the software reaches the actual user.
- Faster Updates: New features and bug fixes can be given to users very quickly.
- Better Teamwork: Everyone in the project group can see if the newly added code breaks the project or works perfectly.

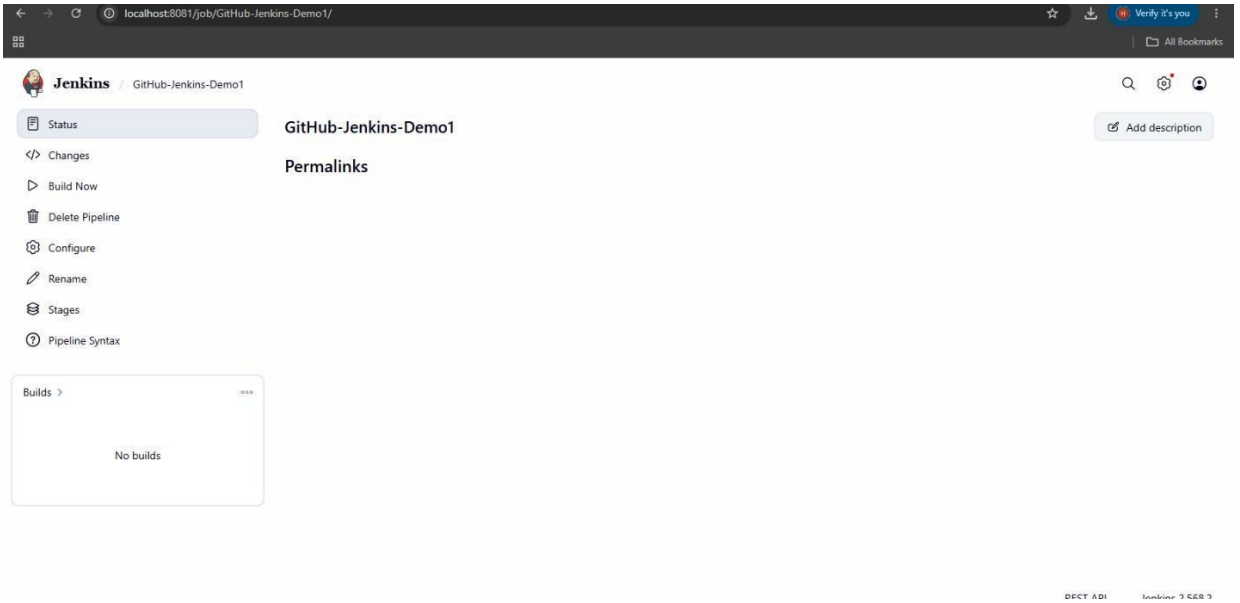
Output:



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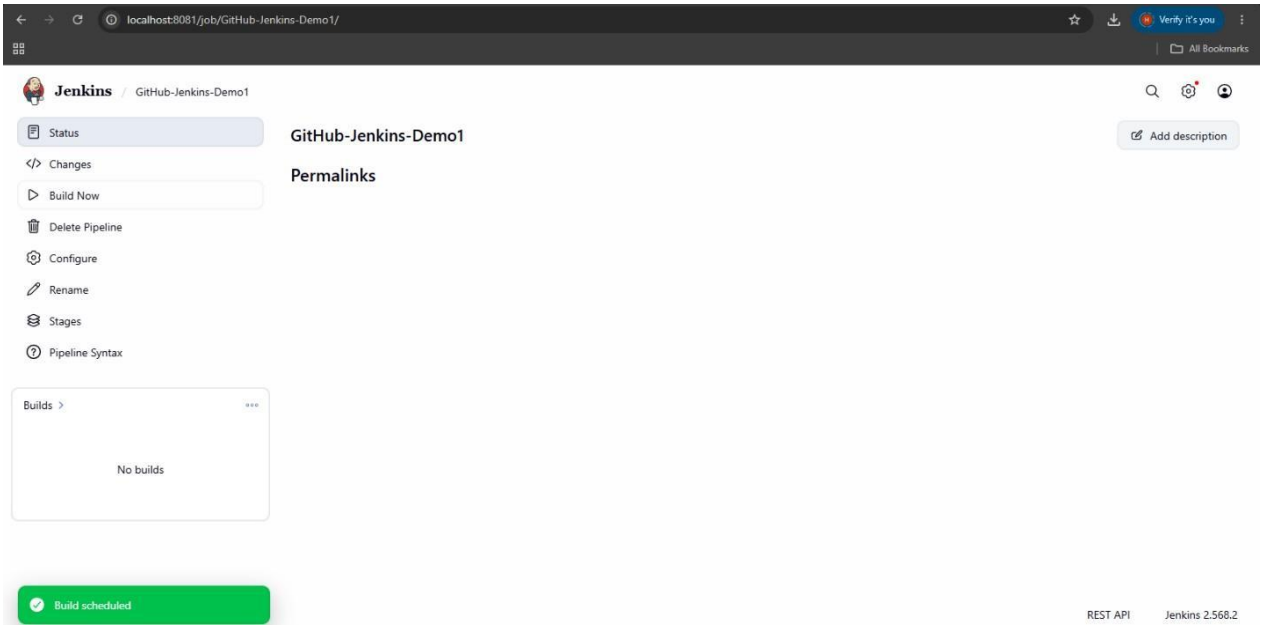
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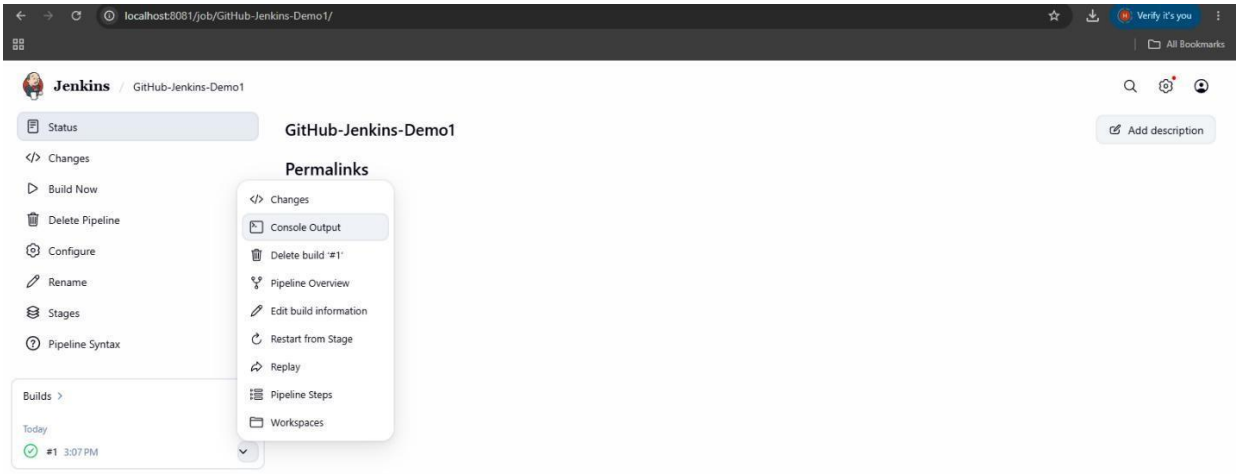




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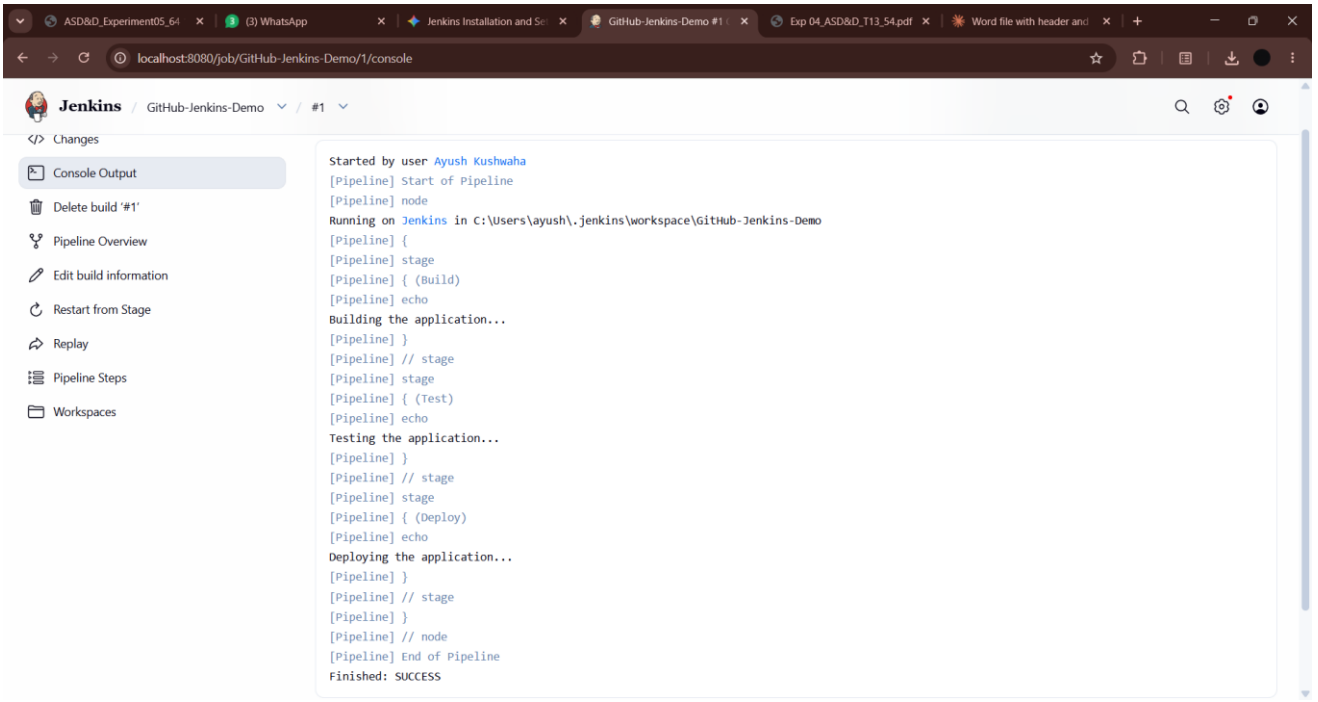
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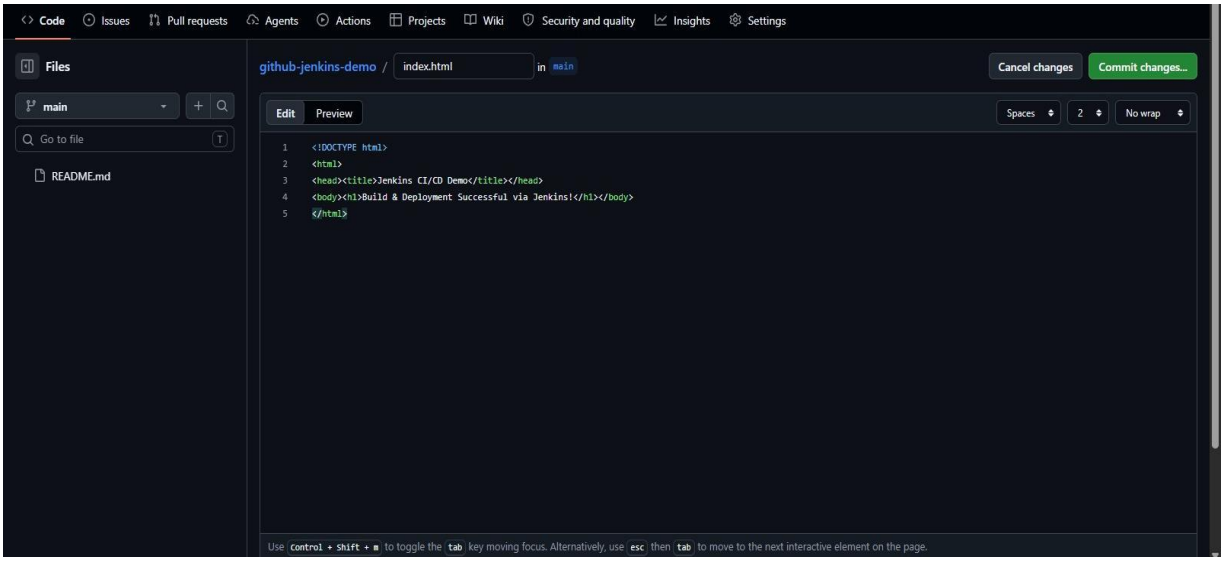




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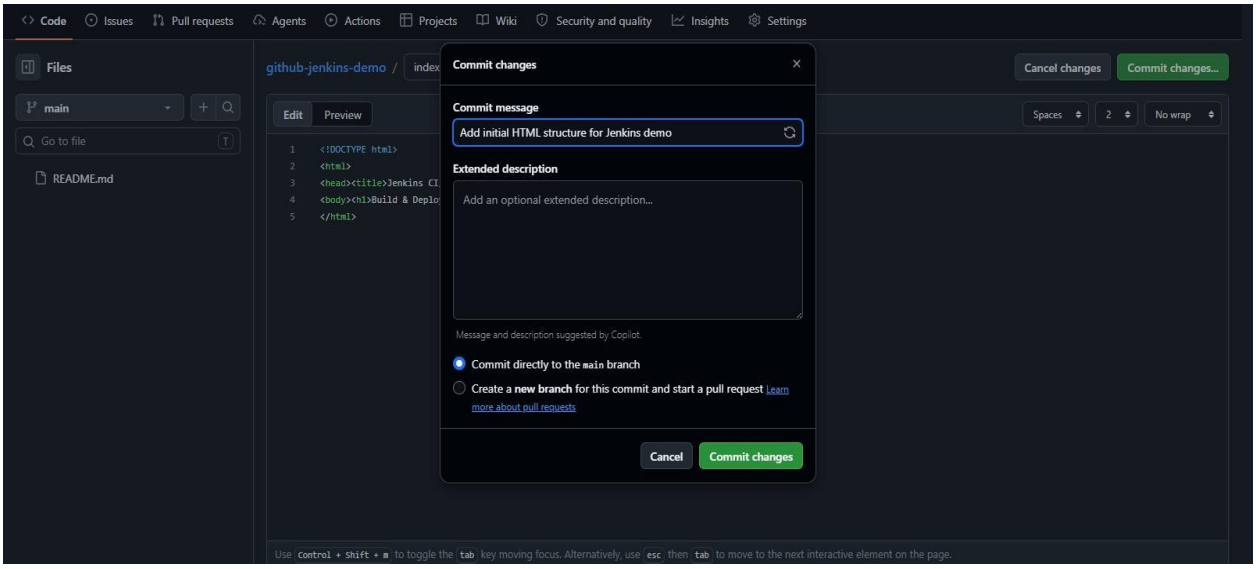


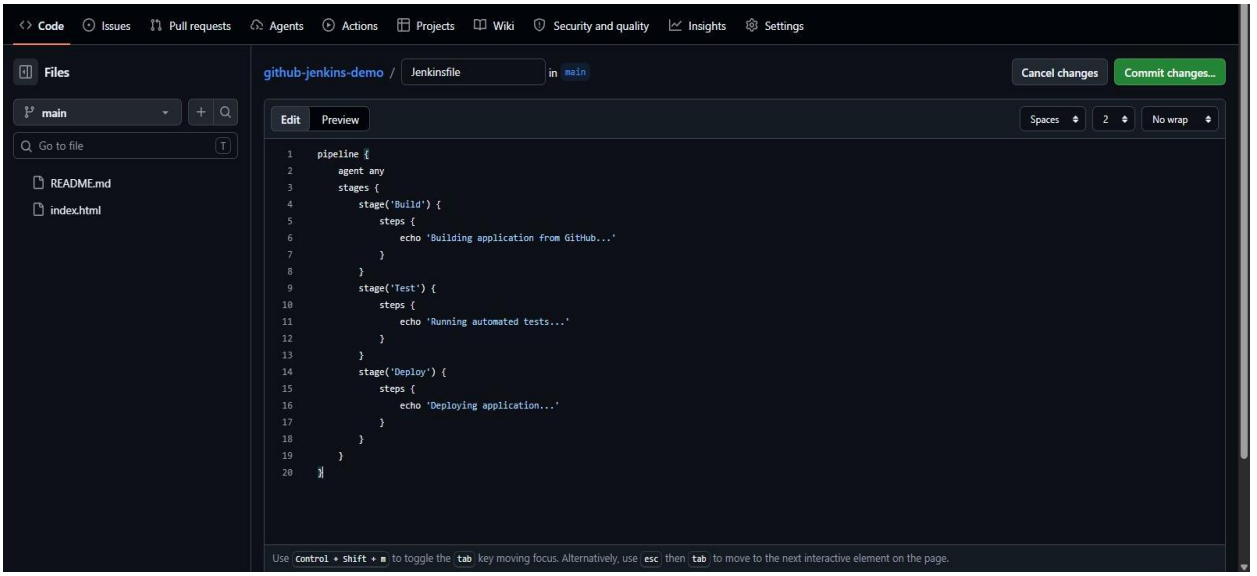
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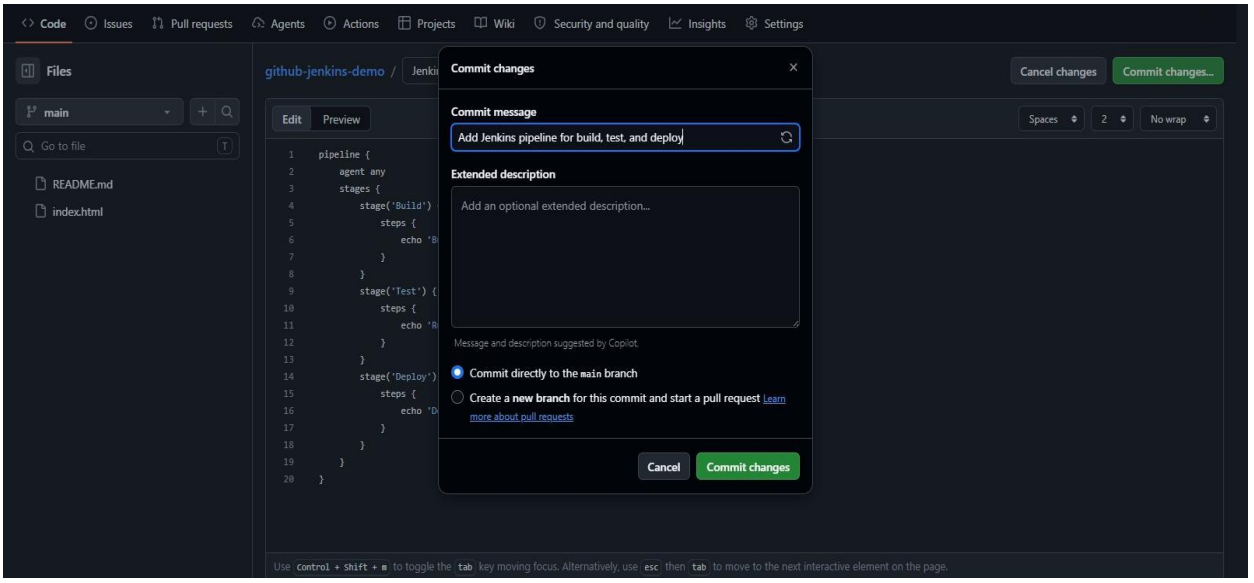
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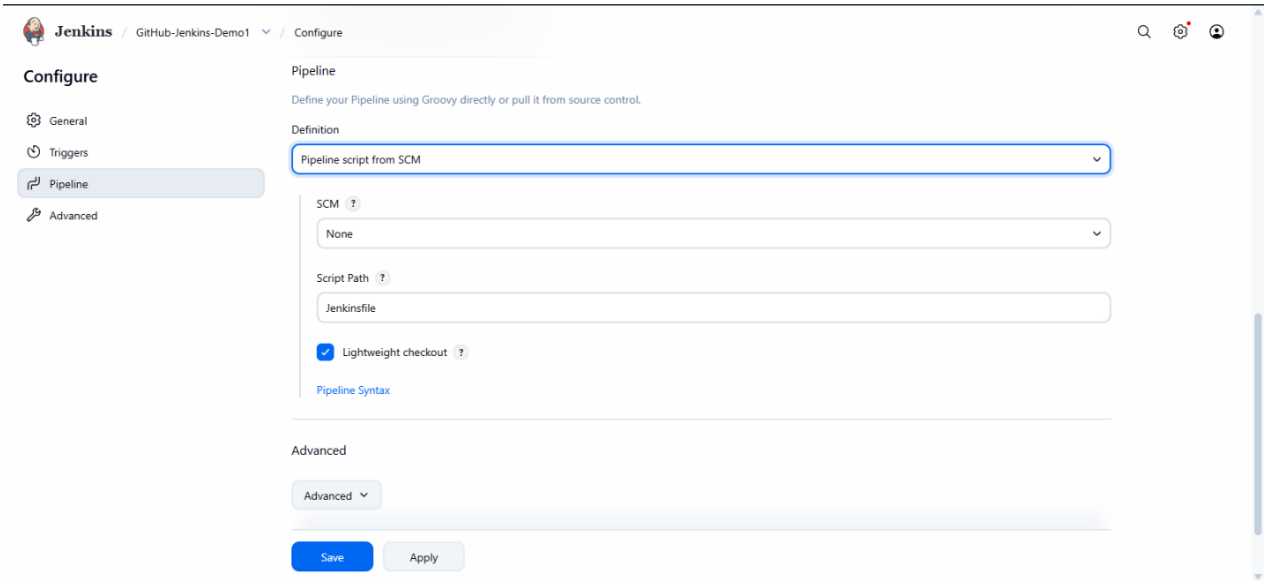
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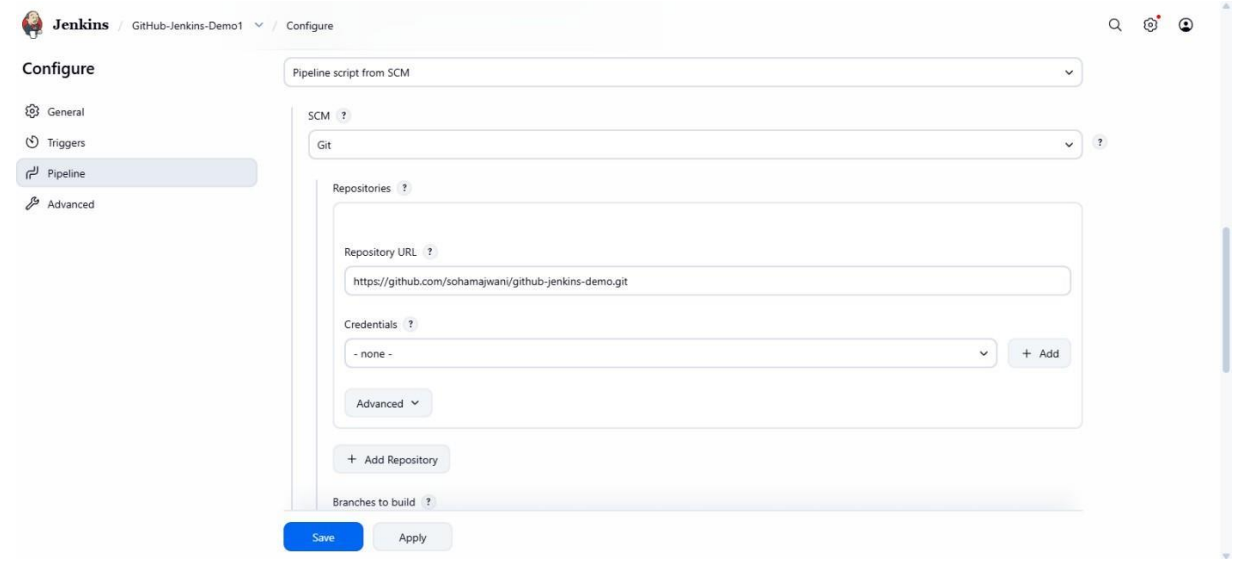




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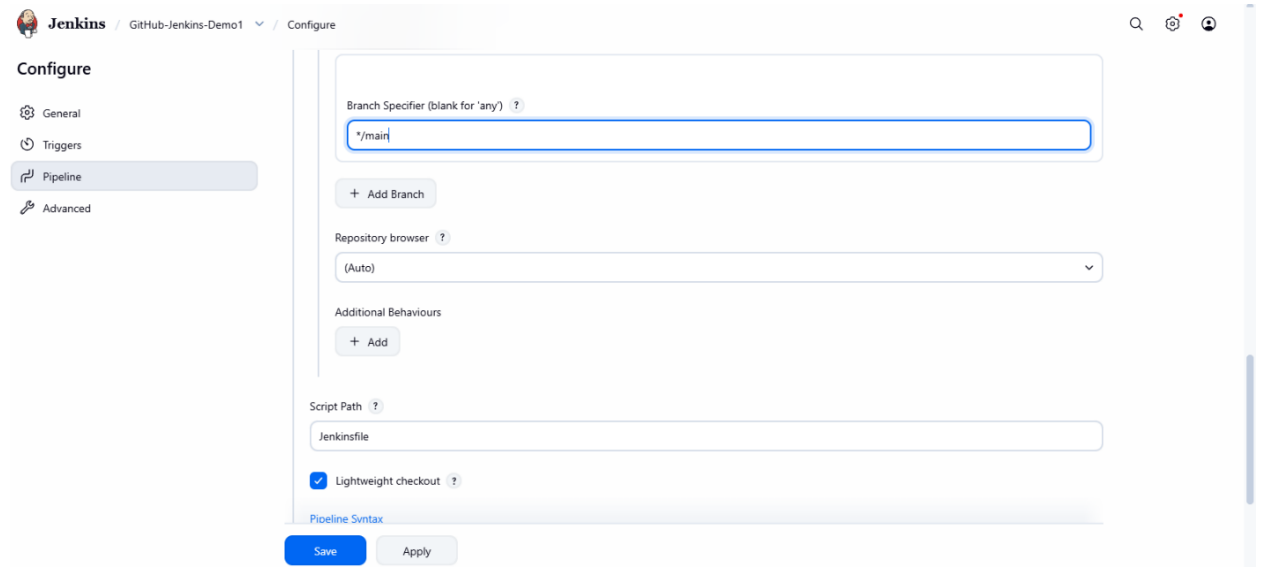
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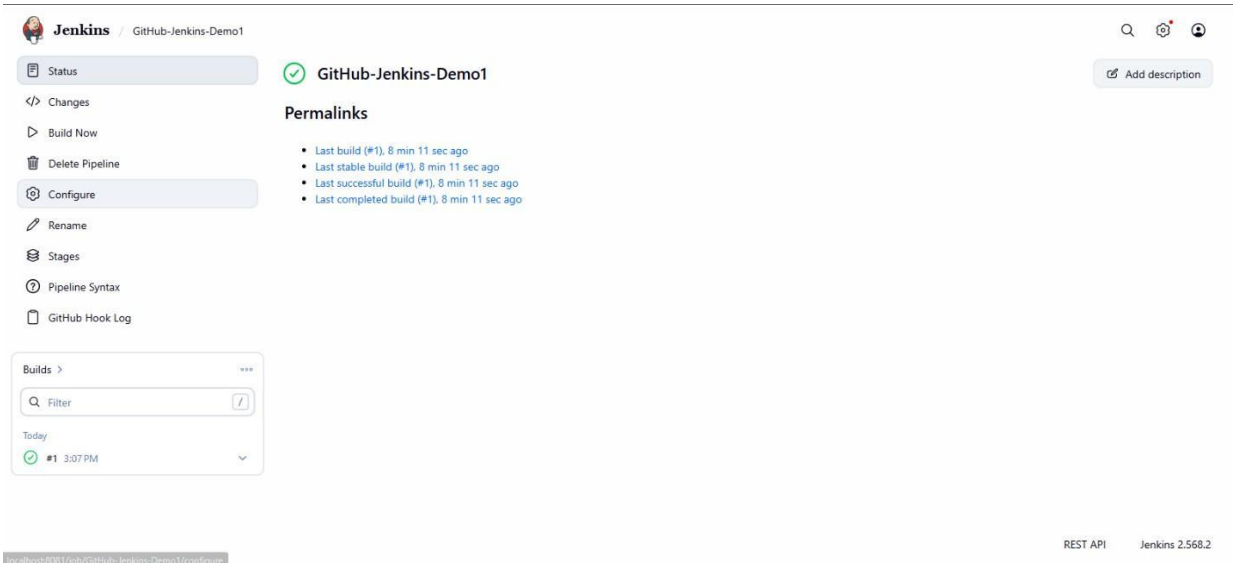




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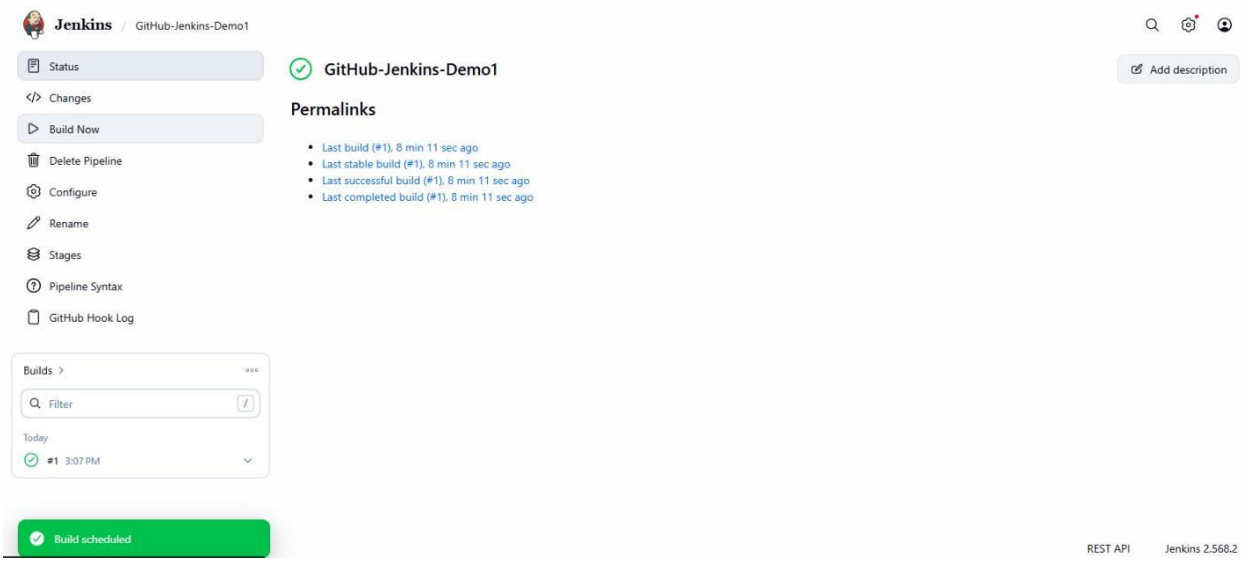
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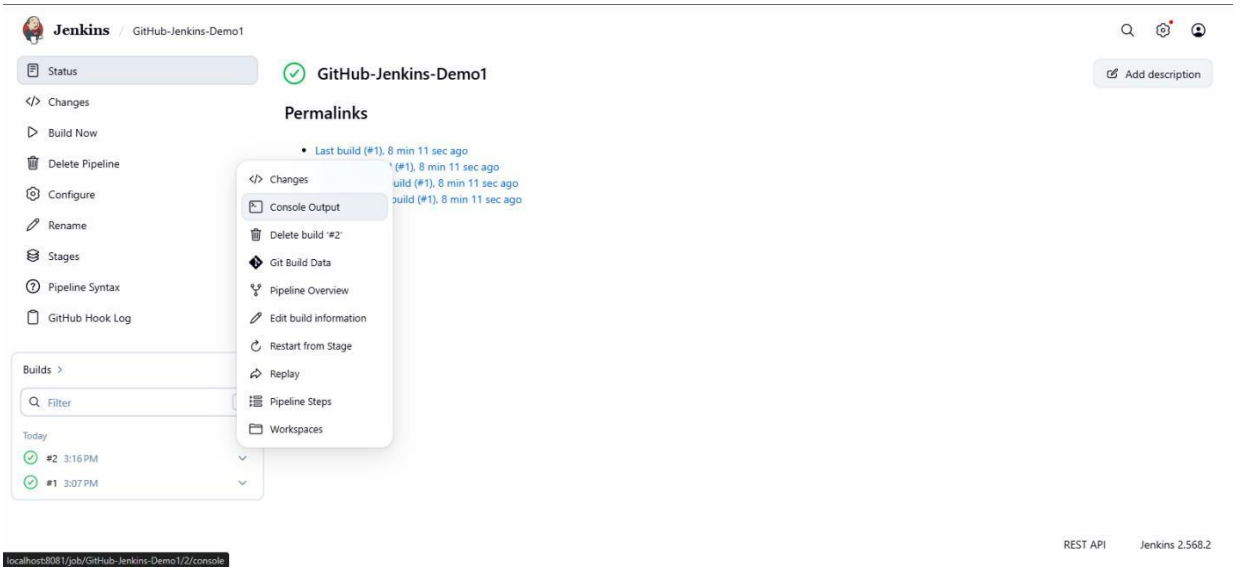




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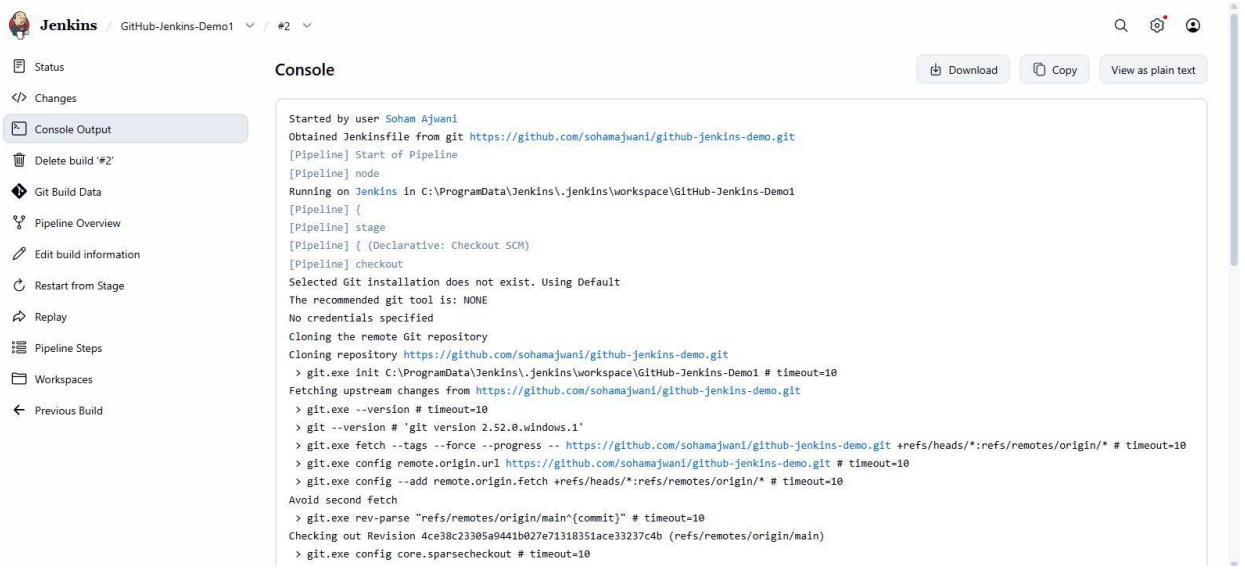
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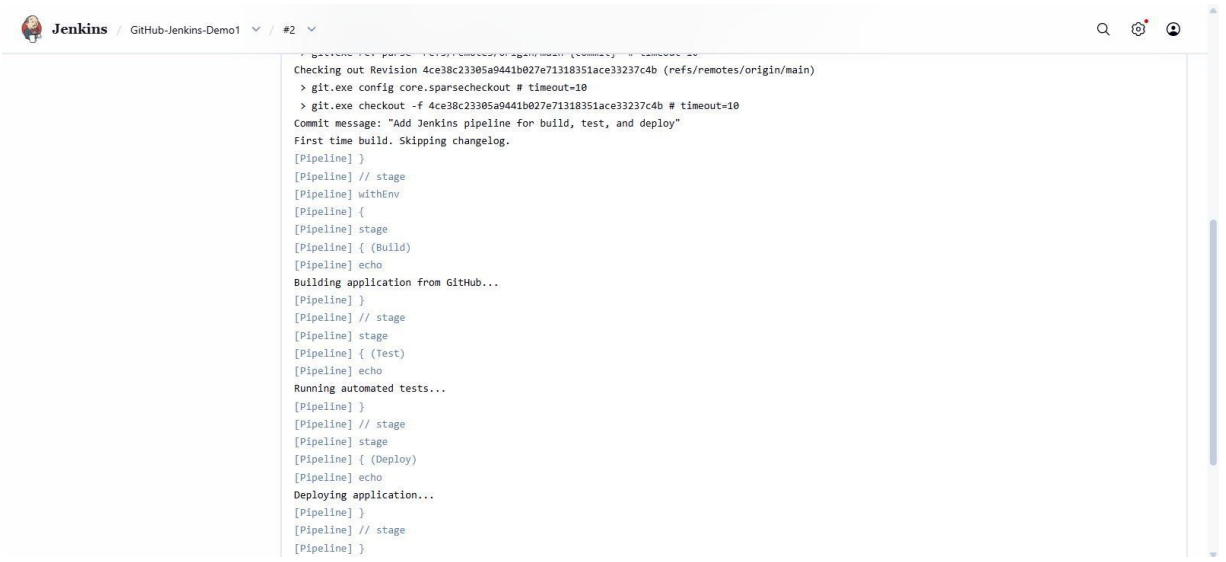




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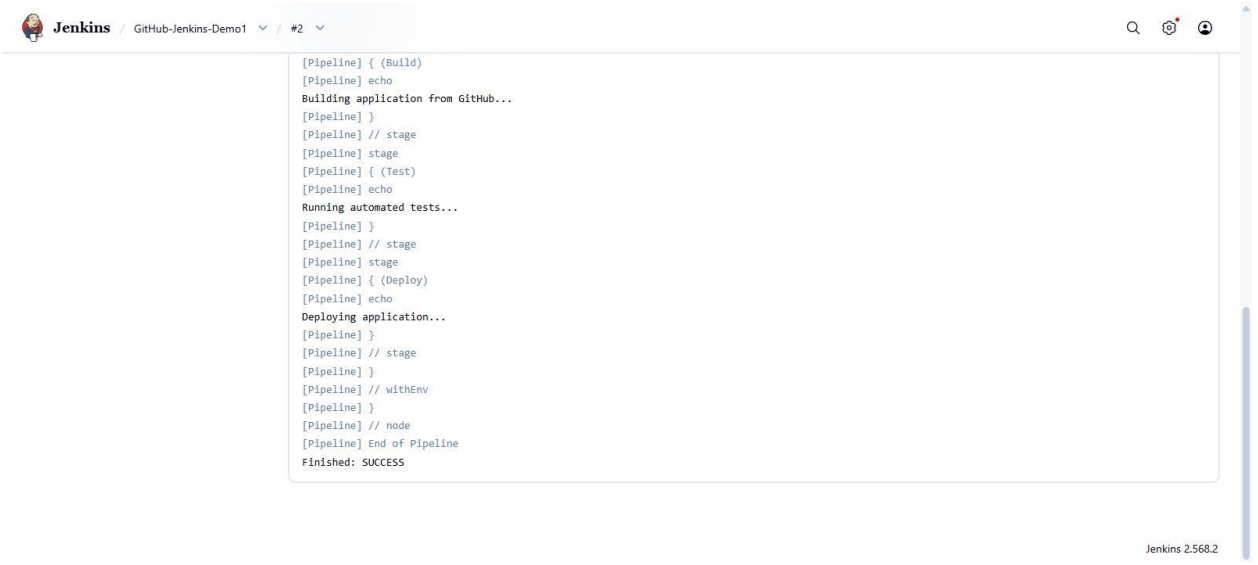
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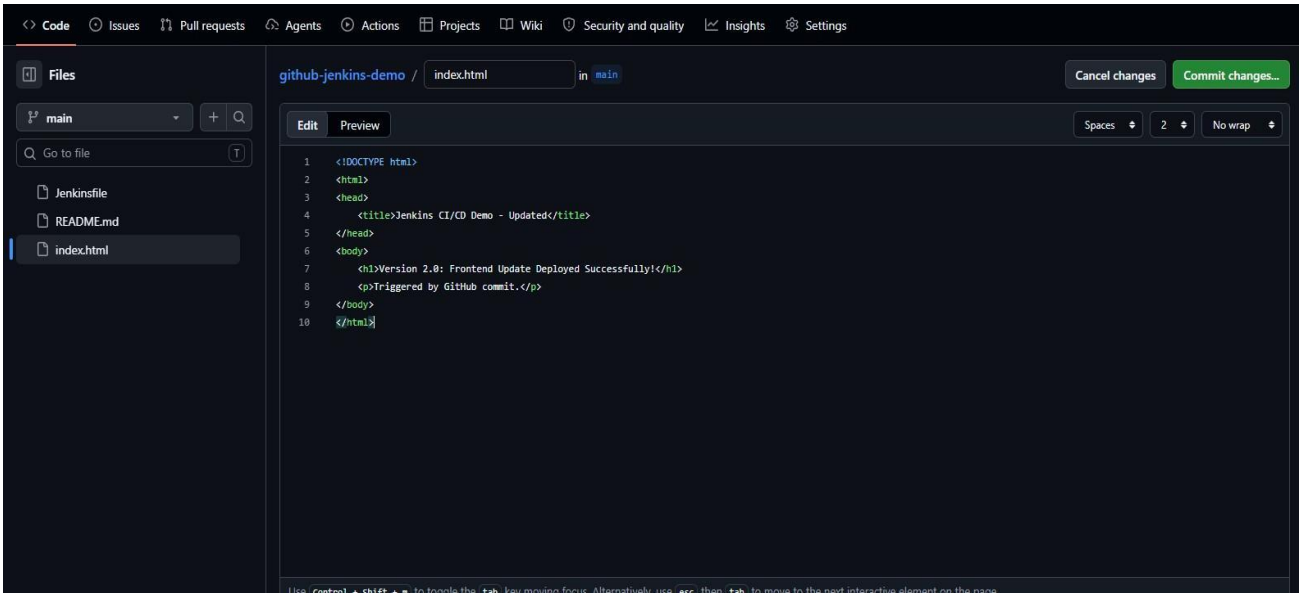




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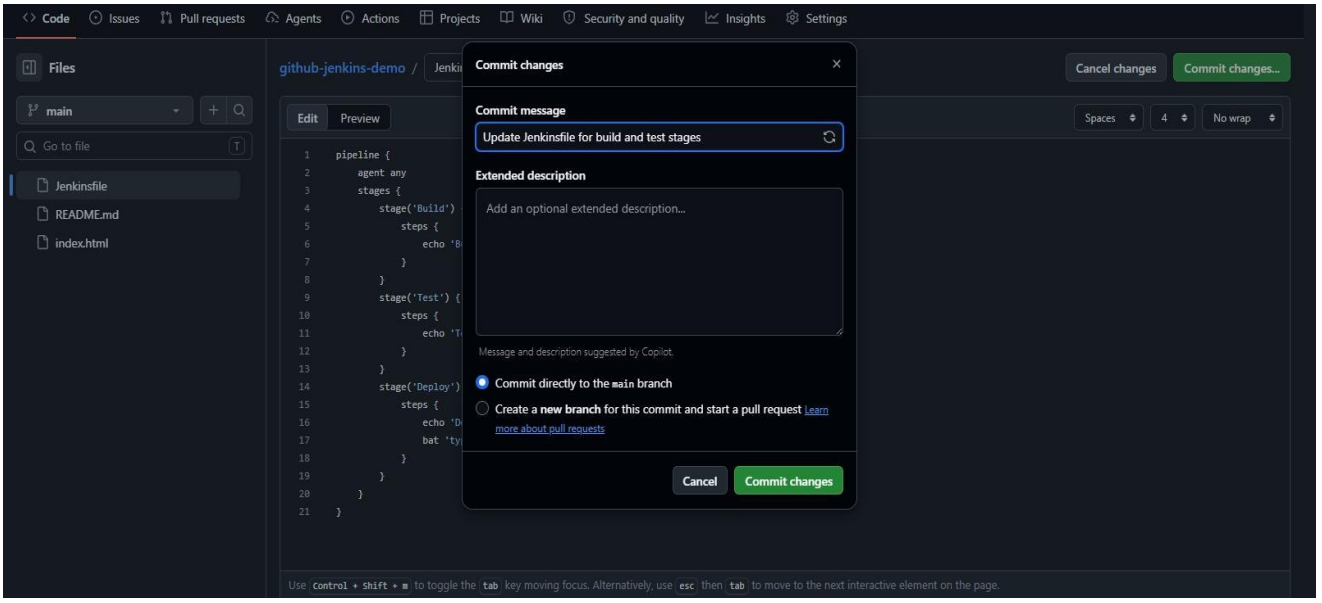
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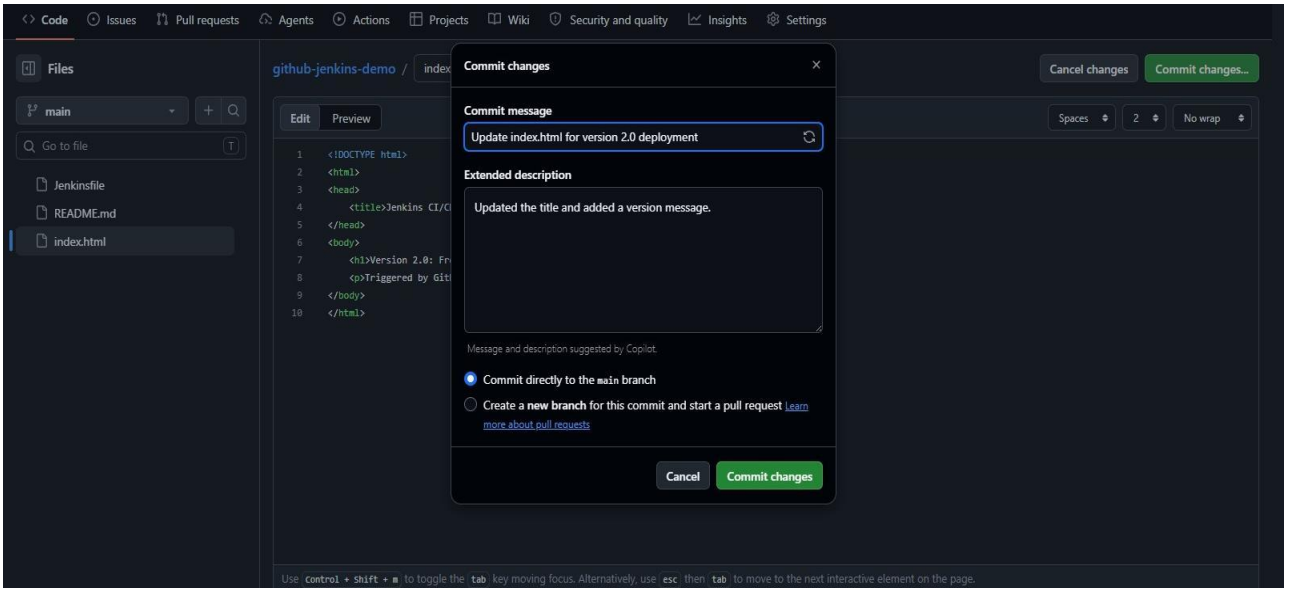




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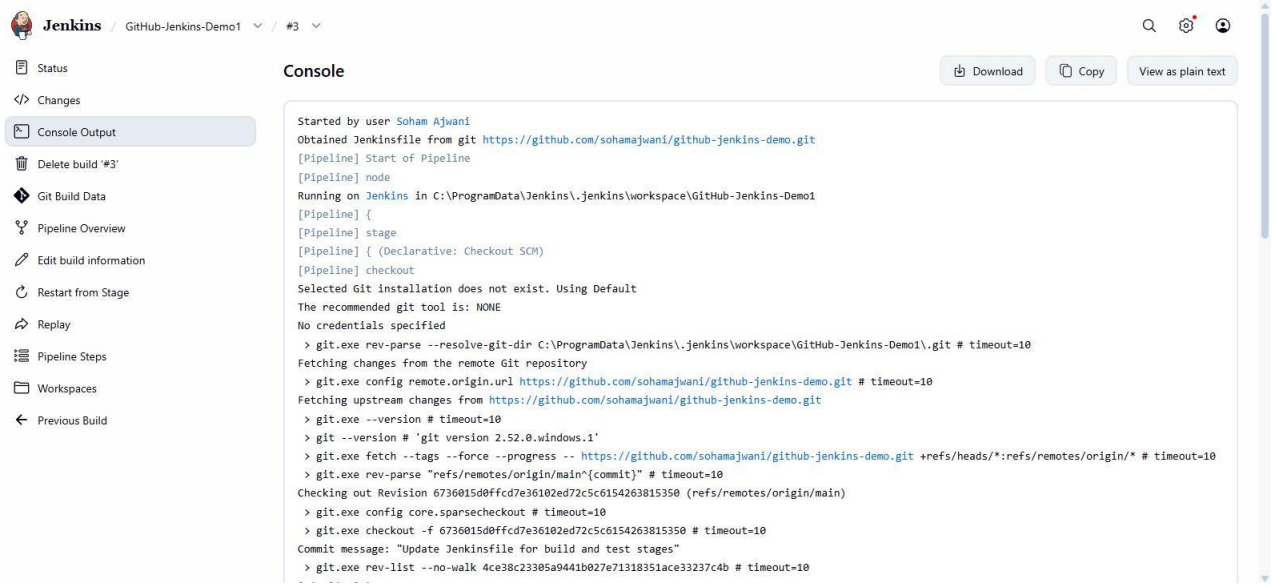
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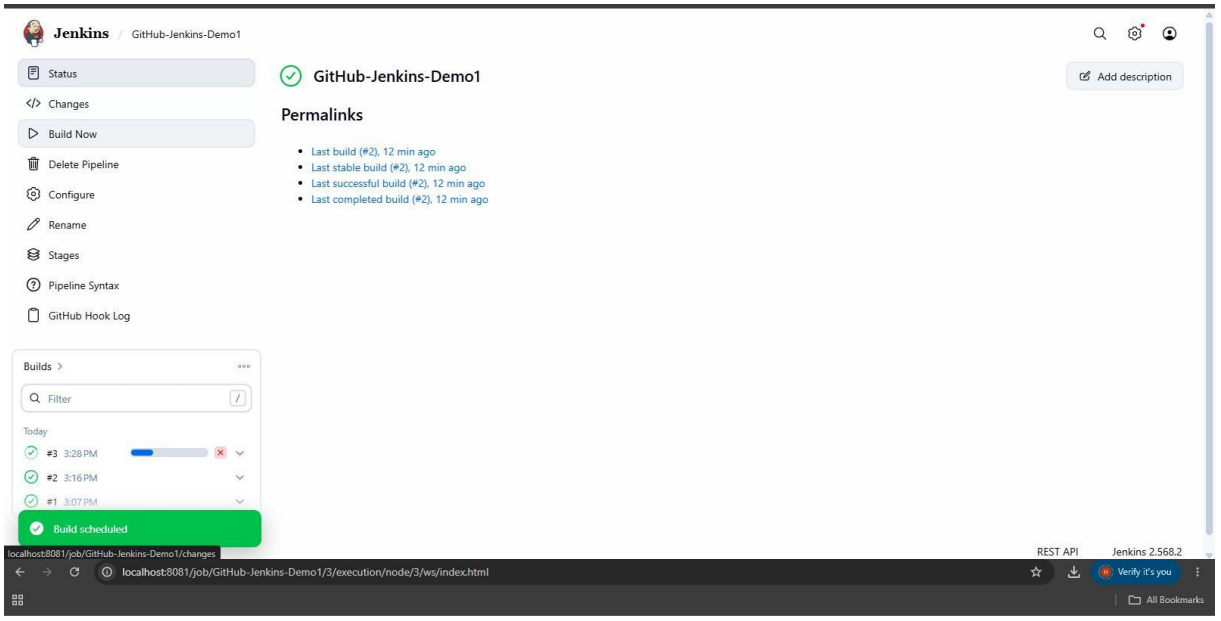
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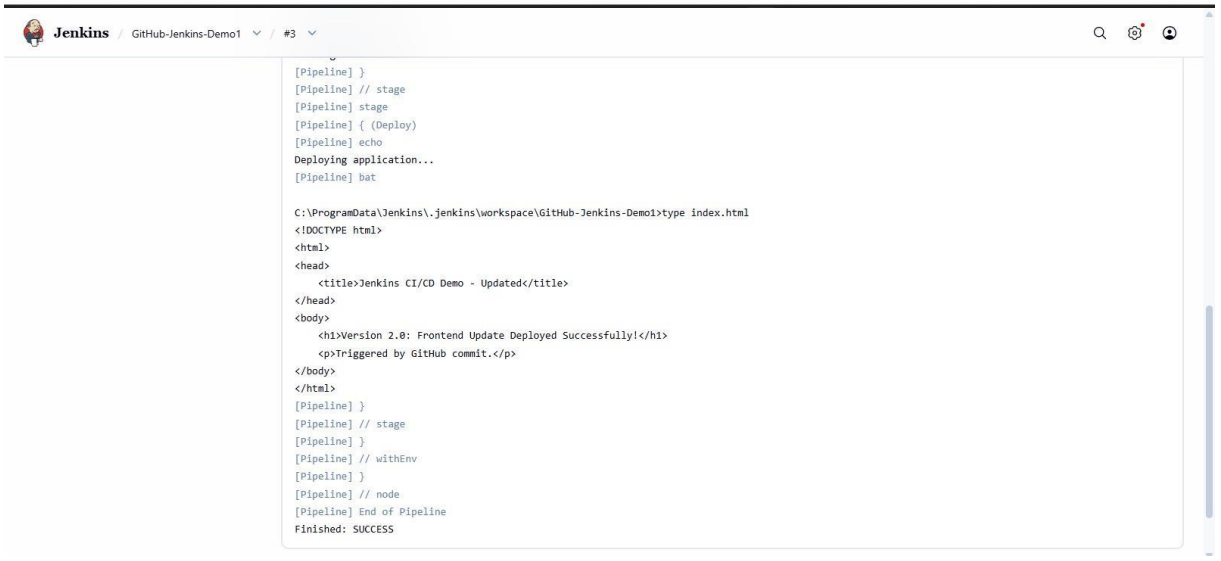
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Version 2.0: Frontend Update Deployed Successfully!

Triggered by GitHub commit.



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Conclusion:

In this experiment, we successfully automated application deployment using Jenkins and GitHub. We learned how to connect a remote GitHub repository to Jenkins using Source Code Management (SCM) and configure a Jenkinsfile to define build, test, and deployment stages as code. Whenever updates were committed to the repository, Jenkins automatically fetched the changes and executed the pipeline, successfully deploying the new version. This practical demonstrated how automated CI/CD pipelines save time, minimize manual deployment errors, and streamline software delivery.